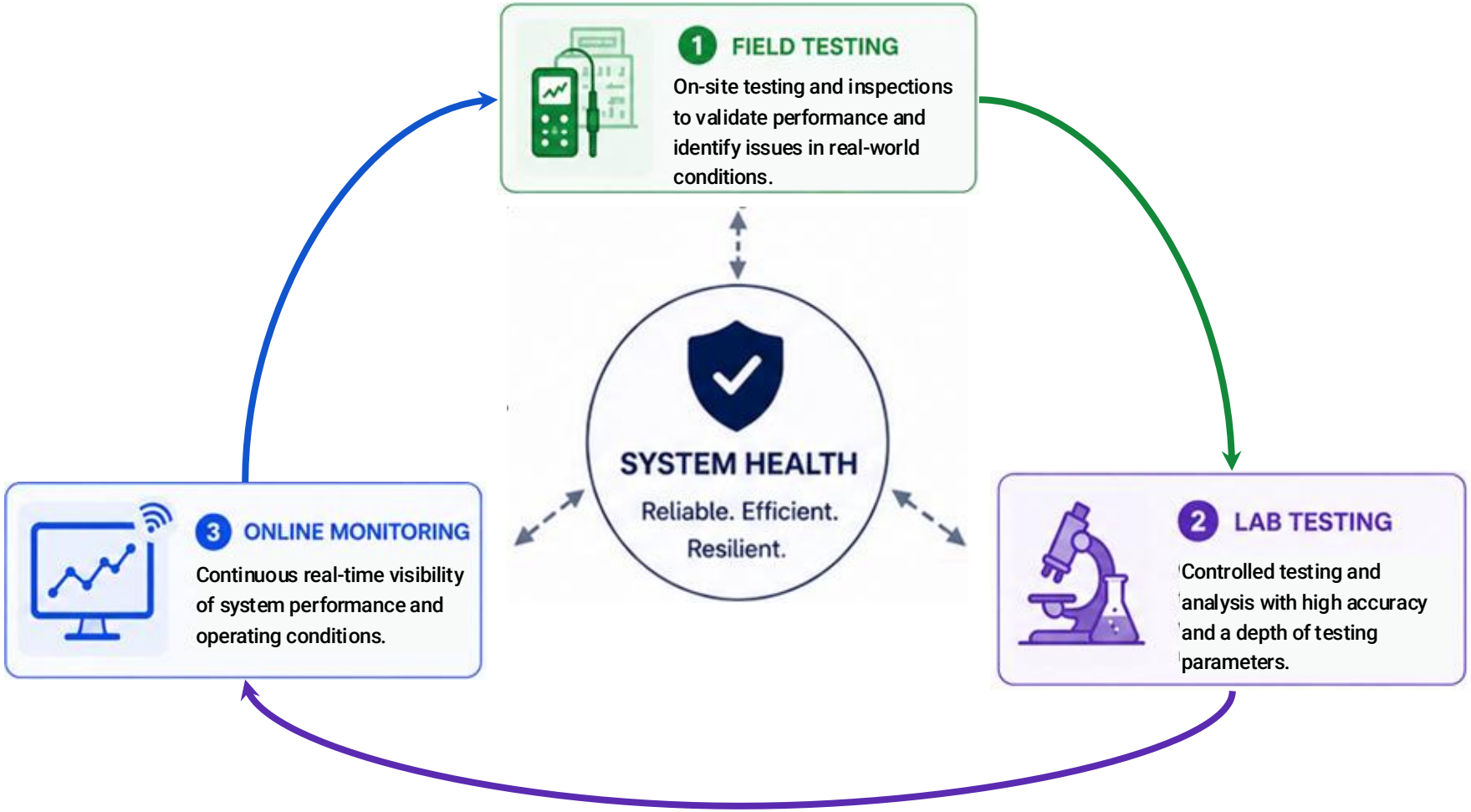
Fluid Monitoring: From Sensors to Insights

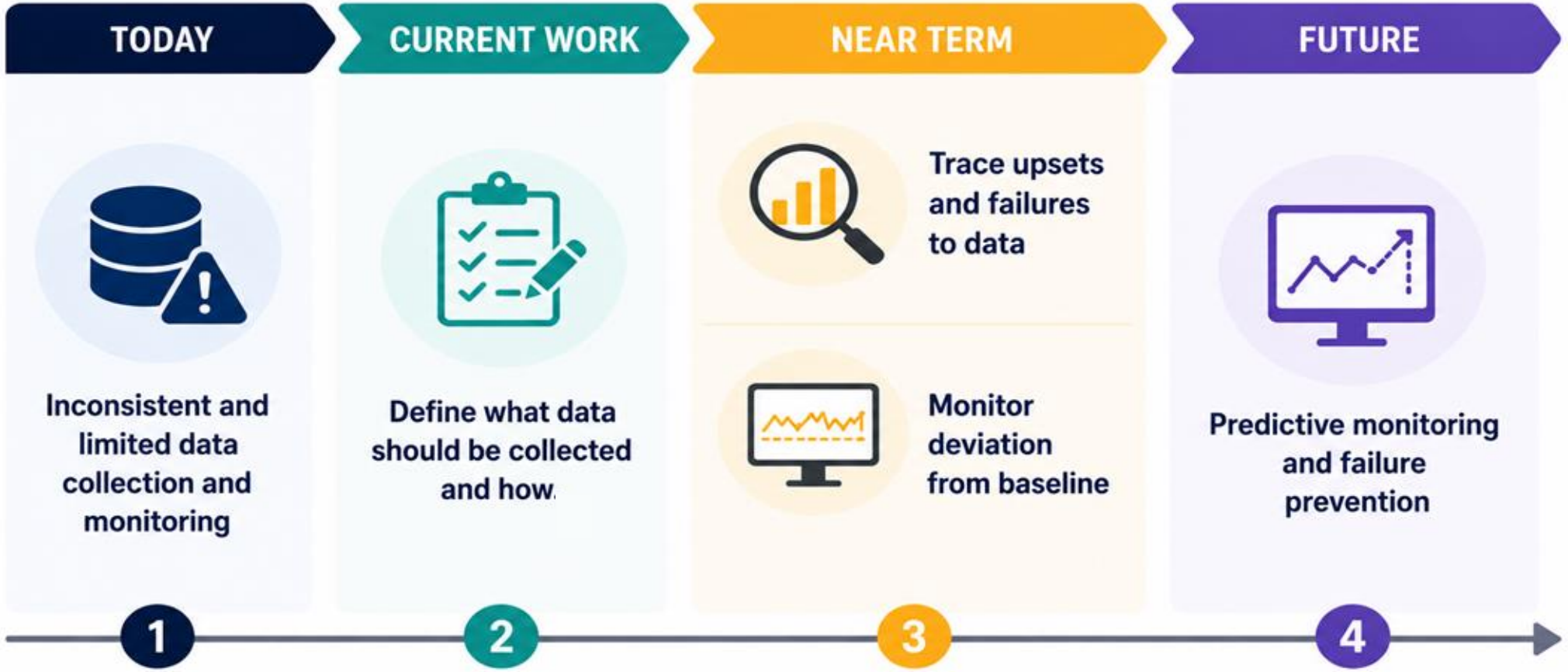
Pascal Michel *CEO - Nordran*
Sreya Dutta *R&D Manager - Dynalene*
Mike Murcia *Corporate Scientist - Ecolab*
Sean Barlett *Principal Availability Engineer - Switch*

OCP BRONZE Member

Sampling Types



Progression of Monitoring



Types of Test Fluids

ID	(PG25% + Water) Coolant Types	Property
RPG	Uninhibited	Control coolant without inhibitor
IAT-PG	Inorganic Acid Technology	Coolant with phosphate for buffering and yellow metal (azole) inhibitor
OAT-PG	Organic Acid Technology	Coolant with organic acid inhibitors for Aluminum, copper and other metals; without phosphate/silicate/borate
LEC-PG	Non-ionic inhibitor package	Coolant with low electrical conductivity < 5 µS/cm that prevents thermal runaway in electronic devices

Note: all tested fluids were undyed

Upset Tests: Contaminant, Inhibitor, Temperature, and Water Effects

Each condition is evaluated separately before coolant properties are compared

TEST 01 / GLYCOL DEGRADATION

Accelerated coolant degradation compared at two temperatures after 450 h*

- 80 °C
- 100 °C

*With and without submerged copper

TEST 02 / CONTAMINANTS

Separate dosing with contaminant ions

- Chloride
- Sulfate
- Hardness

TEST 03 / INHIBITOR LEVEL

Effect of different inhibitor concentrations in IAT coolants

- Phosphate
- Azole (Na-TTA)

TEST 04 / WATER TYPE

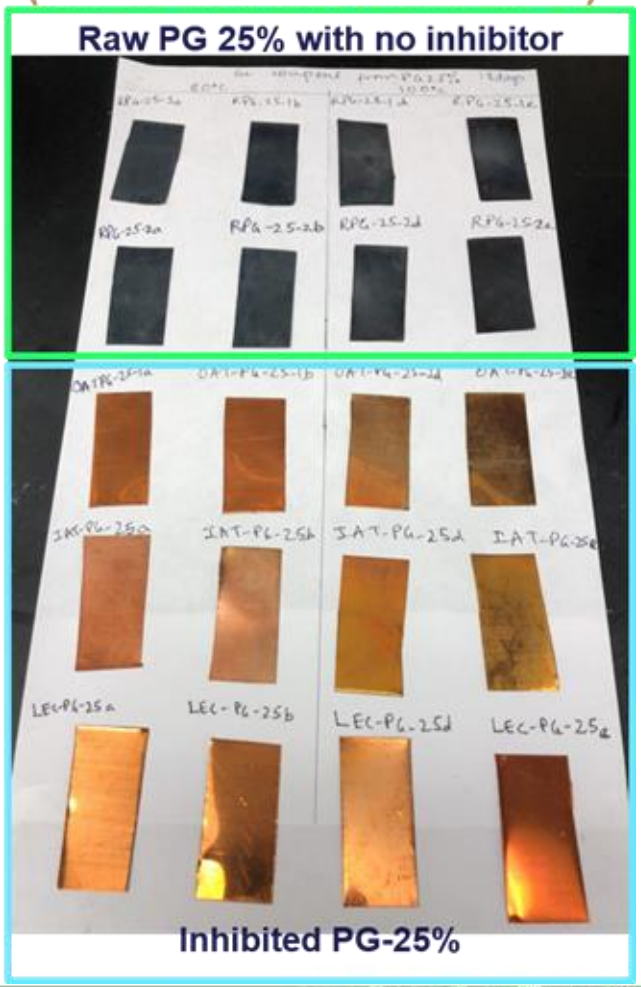
Effect of water source on coolant dilution

- Tap water
- Deionized water

Properties measured: pH, Electrical Conductivity (EC), Reserve Alkalinity (R.A.), Glycol % and Turbidity

Thermal Aging with Copper Results

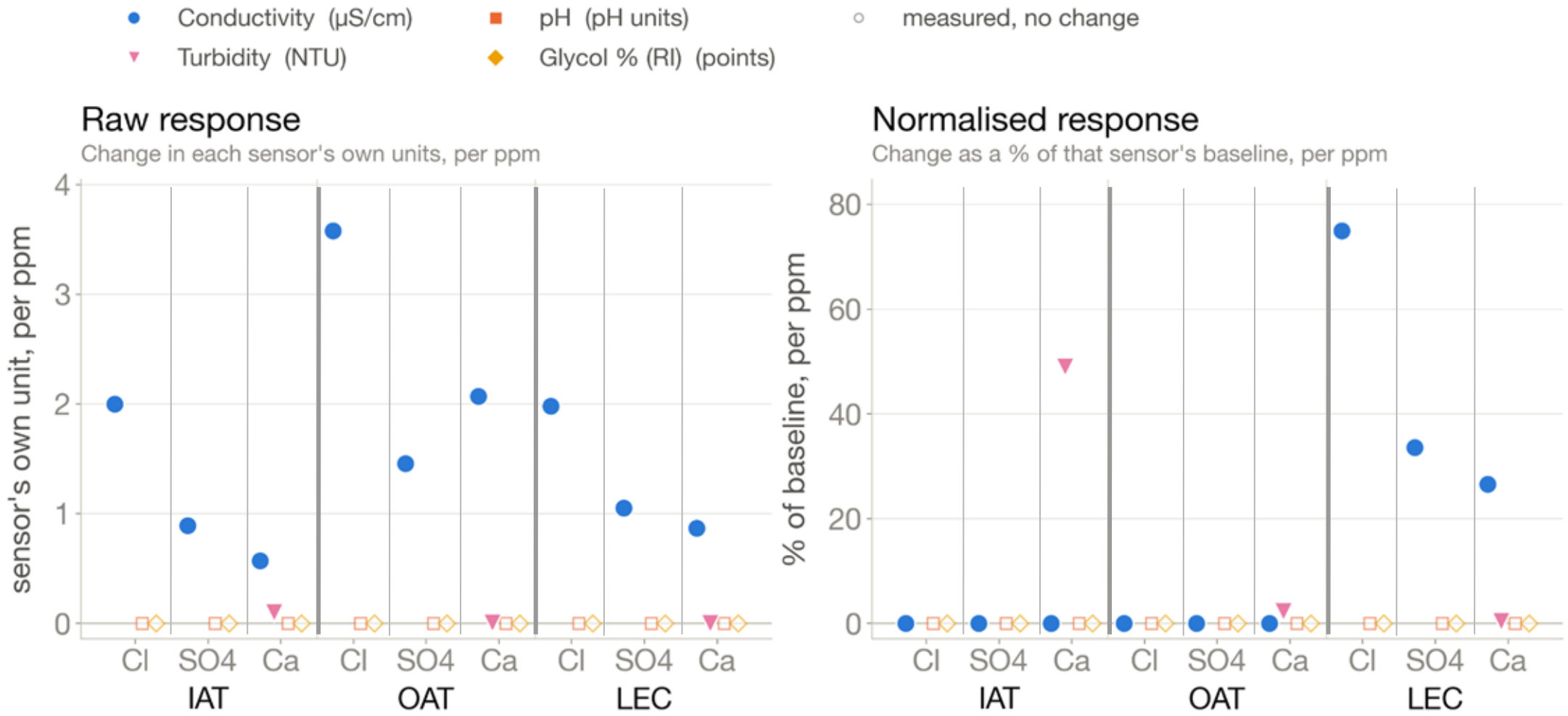
Copper Pieces from Test Coolants (after 450 hours at 80°C and 100°C)



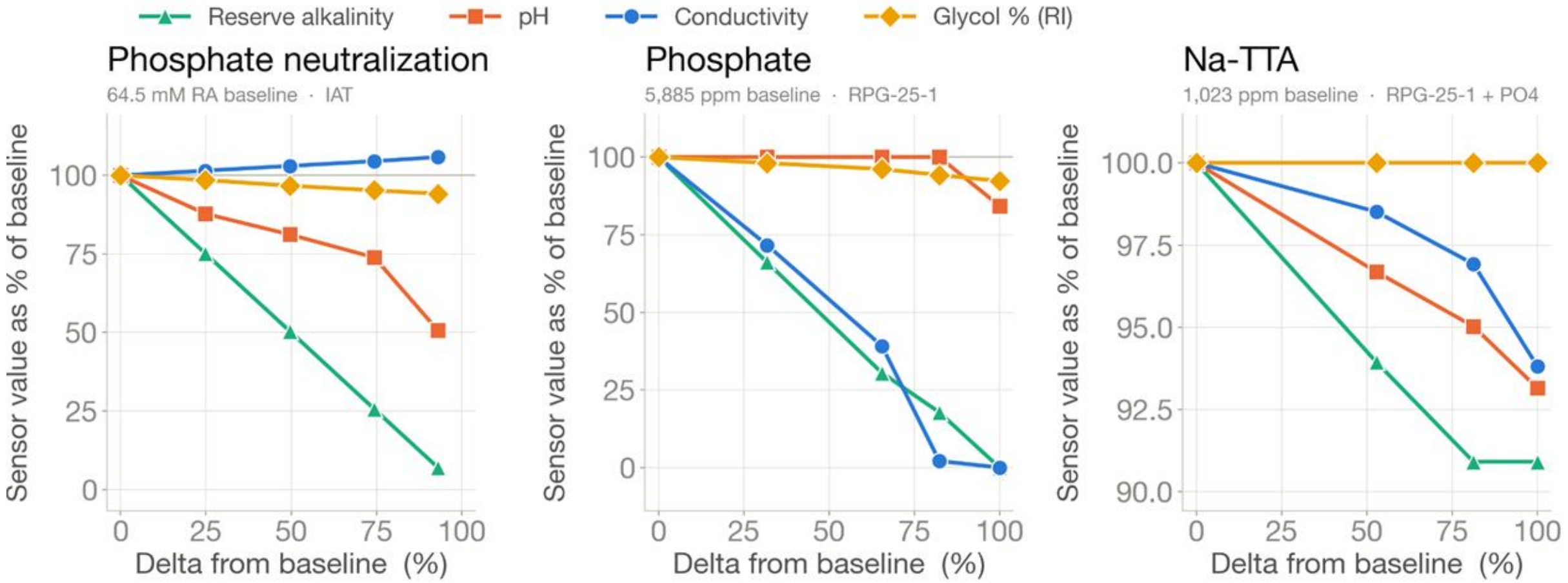
		no copper		with copper	
		80 °C	100 °C	80 °C	100 °C
Acid formed (ppm)	RPG-1	18	1	23	41
	RPG-2	1	14	21	34
Change in pH	RPG-1	–1.24	–1.84	–1.14	–2.04
	RPG-2	–0.11	–1.41	–2.21	–2.51
Copper leached (ppm)	RPG-1	0	0	3	9
	RPG-2	0	0	3	7
Change in EC (µS/cm)	RPG-1	+0.9	+0.9	+8.9	+9.9
	RPG-2	+13.6	+5.6	+29.6	+11.6

Colour is scaled within each metric, from no change to the largest change measured for it.

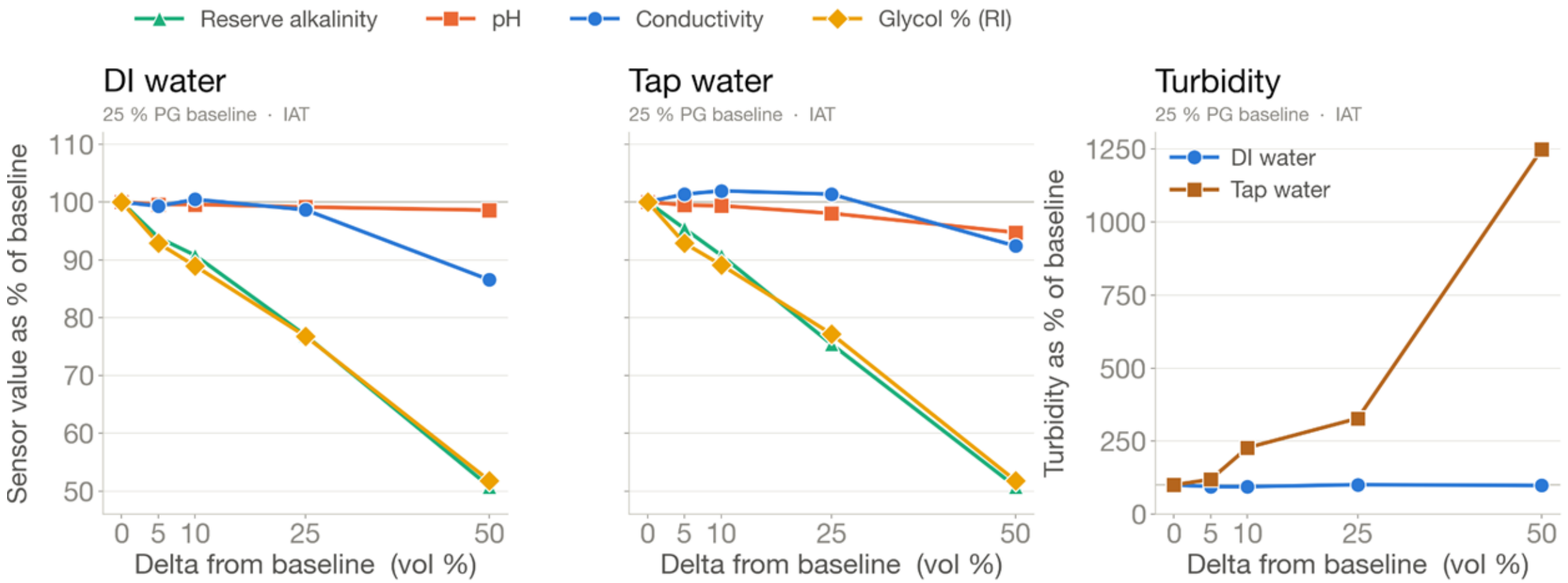
Contamination Results



Inhibitor Levels Results



Dilution Results



Interpretation

The **Univariate Detection Margin (UDM)** helps assess how likely a given sensor could detect a given upset scenario with high confidence and is calculated as follows:

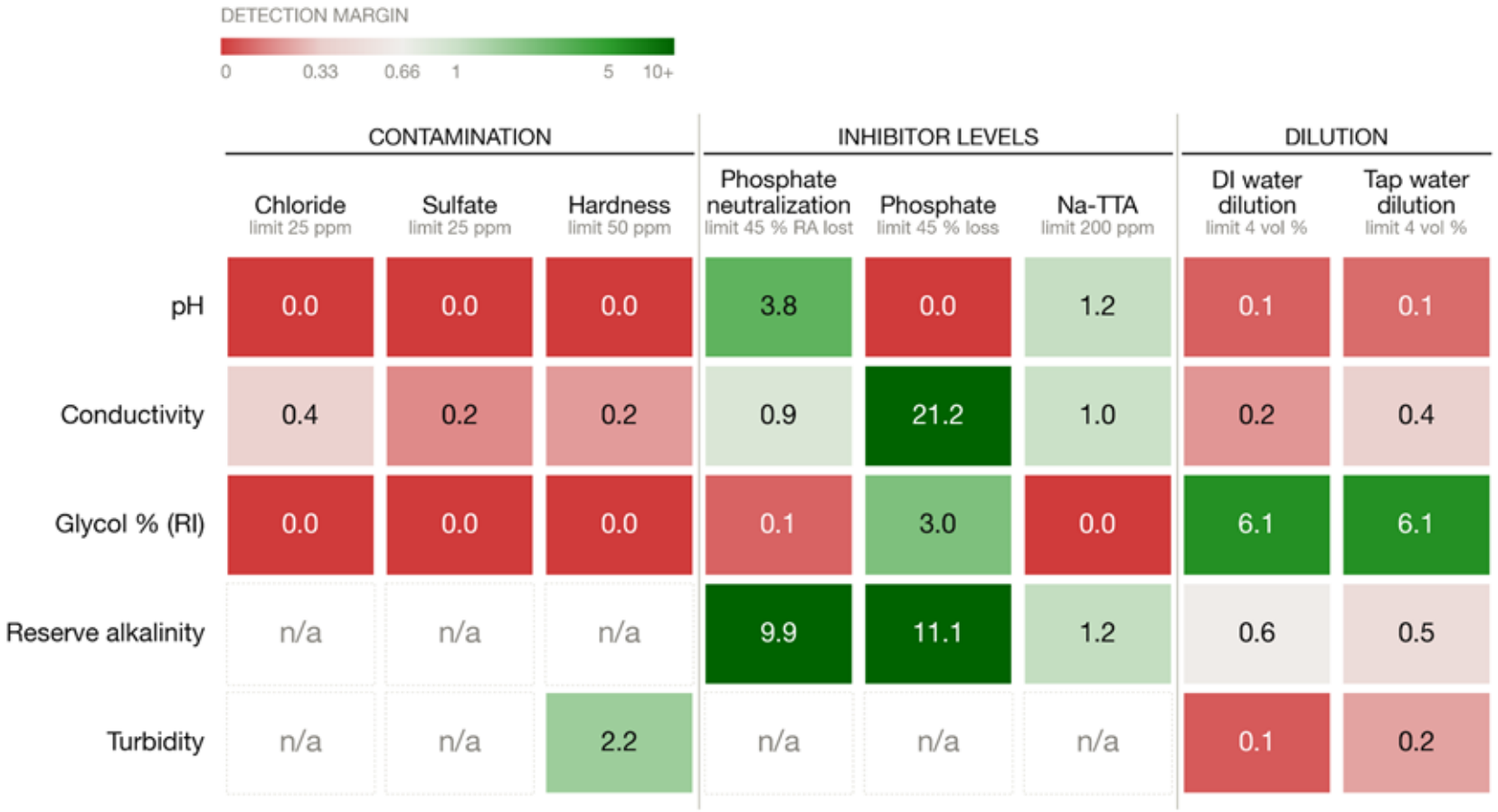
$$\text{Sensor detection threshold (SDT)}: \quad \mathrm{SDT}(T) = \underbrace{3\sigma_{\Delta}(T)}_{\text{stochastic variability}} + \underbrace{D(T)}_{\text{systematic drift}}$$

$$\text{Minimum detectable concentration (MDC)}: \quad \mathrm{MDC}(T) = \frac{\mathrm{SDT}(T)}{\textit{sensor sensitivity in relevant environment}}$$

$$\text{Univariate detection margin (UDM)}: \quad \mathrm{UDM}(T) = \frac{\textit{spec limit by upset}}{\mathrm{MDC}(T)}$$

Note: SDT values are based on sensor testing for all parameters except turbidity, for which it was derived from data in Kahil uoto et al. (2019). Sensor behavior was characterized to determine sensor detection thresholds over a 1-week period.

Univariate Detection Margin - IAT Fluid



	CONTAMINATION			INHIBITOR LEVELS			DILUTION	
	Chloride limit 25 ppm	Sulfate limit 25 ppm	Hardness limit 50 ppm	Phosphate neutralization limit 45 % RA lost	Phosphate limit 45 % loss	Na-TTA limit 200 ppm	DI water dilution limit 4 vol %	Tap water dilution limit 4 vol %
pH	0.0	0.0	0.0	3.8	0.0	1.2	0.1	0.1
Conductivity	0.4	0.2	0.2	0.9	21.2	1.0	0.2	0.4
Glycol % (RI)	0.0	0.0	0.0	0.1	3.0	0.0	6.1	6.1
Reserve alkalinity	n/a	n/a	n/a	9.9	11.1	1.2	0.6	0.5
Turbidity	n/a	n/a	2.2	n/a	n/a	n/a	0.1	0.2

Important Caveats

- Upsets tested were not exhaustive
- Upsets are not all equally likely or equally costly
- Lab tests are not necessarily representative
- Multivariate results may differ from univariate

Univariate Detection Margin - IAT Fluid

Key Takeaways by Sensor for IAT Fluid

- **pH**: can detect acid generation outside buffered regimes and TTA levels
- **Conductivity**: can detect large ion level changes in various forms
- **Glycol % (RI)**: can detect dilution across all fluids
- **Reserve alkalinity**: can detect acid generation, phosphate and TTA level changes, and dilution to a lower extent
- **Turbidity**: can detect hardness

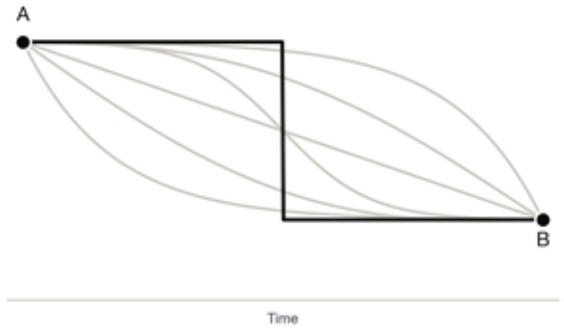
Important Caveats

- Upsets tested were not exhaustive
- Upsets are not all equally likely or equally costly
- Lab tests are not necessarily representative
- Multivariate results may differ from univariate

Multivariate Upset Detection

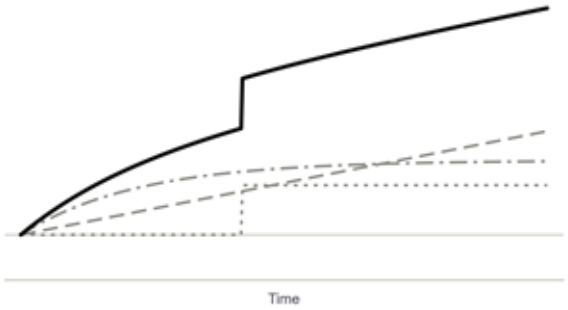
Detecting safe fluid thresholds in real loops with multivariate upsets is both harder and easier

Timeseries



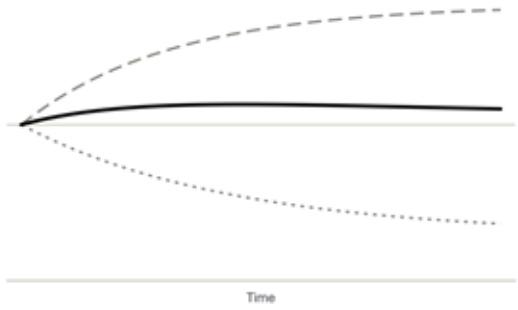
Provides additional context that models can be trained on

Superposition



Makes it difficult to determine how to attribute a sensor reading

Inverse signs



Obfuscates potential large effects that cancel out

Leveraging multiple orthogonal sensors significantly improves multivariate modelling.

Start With the Problem, Not the Sensor

Translating fluid-health risks into practical monitoring packages

1 Fluid-health problem



What failure, upset or perceived risk matters?

Biofouling, corrosion, deposits, dilution, contamination

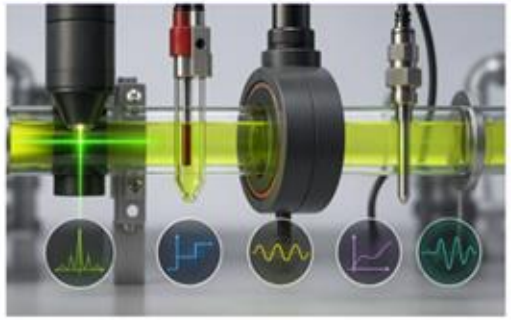
2 Observable fluid change



What changed as the problem developed?

Coolant may show haze, particles or other measurable change

3 Feasible measurement



Can that change be measured continuously and reliably?

Match the change to a practical measurement principle

4 Monitoring package design



How can the measurements be implemented to remain valid and serviceable?

Integrate suitable sensors with representative sampling, data validation, and service access

Once meaningful data are available, use them to:

TREND	Establish a baseline and identify gradual change
INVESTIGATE	Trigger review, confirmatory sampling or root-cause analysis
MAINTAIN	Support condition-based fluid or system maintenance

Effective monitoring prevents fluid-health problems through feasible measurements driving actionable insights

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Match the Sensor Technology to the Measurement

CORE MEASUREMENTS

Measurement	Value provided	Preferred sensor attributes	Critical qualification
pH	Indicates coolant acidification or biological activity and increasing corrosion risk	**Protected industrial glass electrode; gel-filled reference; automatic temperature compensation**	Calibration-dependent; provide access for verification and planned replacement
Conductivity	Detects dilution or other ionic change associated with contamination, additive loss or reduced biostatic protection	**Toroidal or inductive sensor; temperature-compensated**	Confirm adequate sensitivity across coolant range; interpret relative to the formulation baseline
Refractive index	Measures glycol concentration to detect dilution and loss of formulation or biostatic protection	**Inline optical refractometer; temperature-compensated**	Validate correlation for the actual coolant formulation; other dissolved species may affect RI
Temperature	Enables accurate compensation and comparable trends for supported measurements	**Stable, linear Pt1000 RTD; 4-wire preferred where practical**	Co-locate with compensated sensors and measure the same active fluid sample
Turbidity	Detects suspended or colloidal material associated with contamination, corrosion products or biofilm	**Low-maintenance optical sensor with suitable range and stable optical path**	Bubbles, coolant color, deposits and window fouling may affect response
Sample flow	Validates exposure to a fresh, representative sample	**Reliable low-flow meter or flow switch; quantitative where flow affects performance**	Inadequate or unstable sample flow should make associated data invalid, not normal

SUPPLEMENTAL / APPLICATION-SPECIFIC

Reserve alkalinity	Corrosion ER / LPR	Particle counting / imaging	Ion-selective electrodes	Dissolved oxygen	Deposit / fouling sensors
Buffer capacity	**Probe-surface corrosion**	**Size distribution or morphology**	**Specific ionic species**	**Air ingress or loop integrity**	**Deposition tendency**
Titration; periodic offline analysis	May not represent the full loop	Flow, bubbles and fouling	Selectivity, drift and calibration	Calibration and maintenance	No broadly accepted TCS method

Consider when added insight justifies added maintenance, deployment, interpretation or standardization complexity.

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Sensor Implementation

Centralize Monitoring at a Representative, Serviceable Location



CDU-INTEGRATED MONITORING

Engineered into New CDU Equipment



STANDALONE MONITORING

Engineered Retrofit Applications

Representative sample

Continuously refreshed fluid representative of overall closed-loop coolant condition.

Centralized service and sample access

One location for inspection, calibration, replacement, grab samples and at-line or offline verification.

System-specific hydraulics

Available pressure and differential pressure influence sample-loop and sensor-package design.

Separate from control instrumentation

Fluid quality complements distributed temperature, pressure and flow instrumentation used for thermal and hydraulic control.

CONSIDER REDUNDANCY WHEN MEASUREMENT CONFIDENCE IS CRITICAL

Use independent at-line or offline verification, duplicate online measurement, or multi-sensor consensus in proportion to the consequence of being wrong.

Group-controlled CDUs: preserve monitoring availability during CDU rotation or service.

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Key Takeaways

1. The optimal online measurement set is still evolving
2. Individual sensors can already detect meaningful upsets
3. Holistic detection requires multiple measurements + models
4. Deployment is a core part of sensor performance
5. Coolant type determines which parameters matter most, so multi-parameter monitoring enables earlier upset detection.

Call to Action

- Join the Cooling Environments - Fluid Health and Coolant Monitoring Workstreams
- Timeline for contribution availability -
  - Guidelines for Real Time Monitoring of PG Fluids - Q4 26
  - Measuring upsets in PG Fluids - Q1 27
- Where to find additional information (URL links)
  - Project Wiki: https://www.opencompute.org/wiki/Cooling_Environments/Cold_Plate

Open Discussion

